

9. Cluster List

- The clusters are firstly created based on the reflected signals of the subjects. Each particle will be detected as clusters. The nearest particle will be the first cluster that will be detected and taken to the interface. With a rising number of clusters the distance rises, too.
- Later clusters will build objects which is more useful.
- The Cluster list output consists of up to three message definitions that are sent in a regular interval (about 70 to 80 ms) if clusters are selected in signal **RadarCfg_OutputType** (0x200).
 - Cluster_0_Status** (0x600) – The first message contains list header information, i.e. the number of near scan clusters and number of far scan clusters that are sent afterwards.
 - Cluster_1_General** (0x701) – This message contains the position and velocity of the clusters and is sent repeatedly for all the detected clusters (first near scan, then far scan). Each of the two cluster lists is range sorted. If there are more than 250 clusters, only the first 250 clusters are sent.
 - Cluster_2_Quality** (0x702) – This message contains the quality information of the clusters and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all clusters in the same way as message **Cluster_1_General** (0x701).

If the quality information is sent, first all messages of type **Cluster_1_General** (0x701) are sent and afterwards all messages of type **Cluster_2_Quality** (0x702).

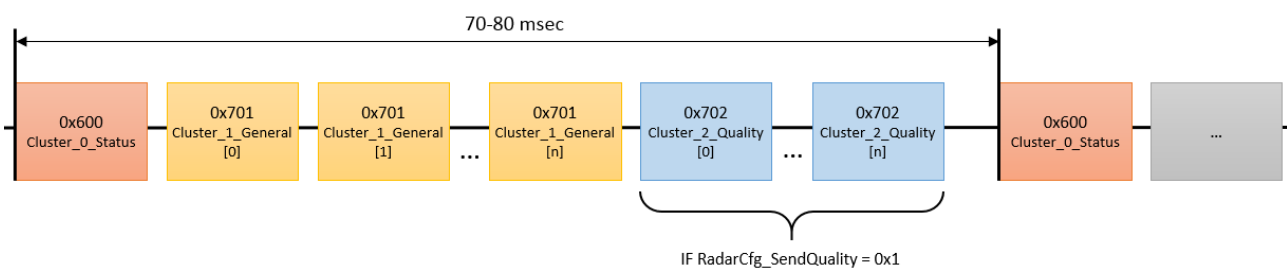


Figure 24: Overview of cluster list messages that are sent cyclically by the radar.

9.1. Cluster list status (0x600)

The message **Cluster_0_Status** (0x600) contains the cluster list header information and is sent as first message of the cluster list output and only once per measurement cycle.

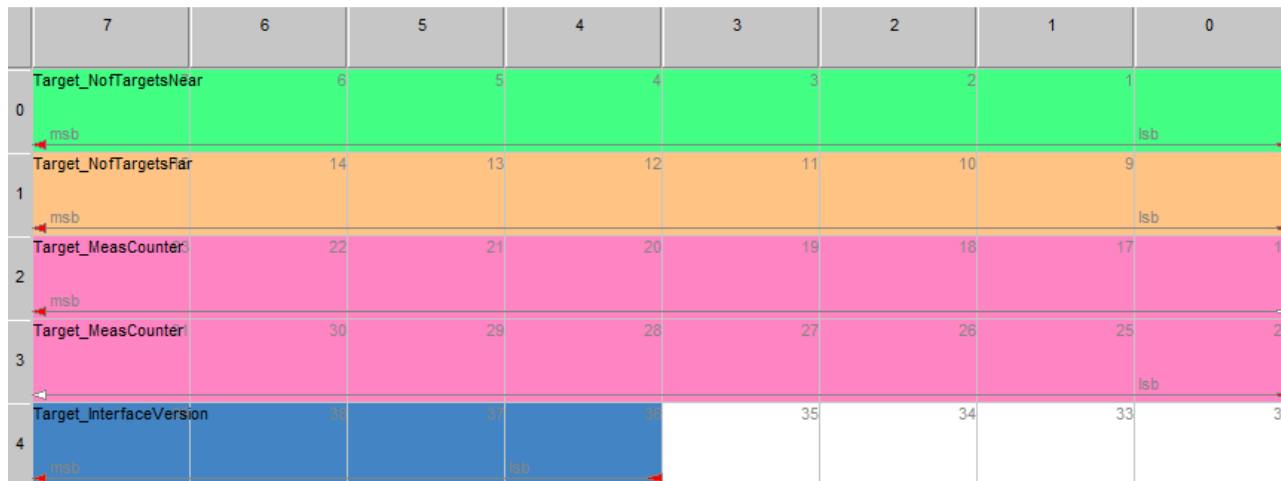


Figure 25: Cluster_0_Status - message layout (0x600)

Table 29: Cluster_0_Status - message content (0x600) [ARS 404-21](#)

Signal	Start	Len	Min	Max	Res	Unit
Cluster_NofClustersNear	0	8	0	255	1	-
Cluster_NofClustersFar	8	8	0	0		
Cluster_MeasCounter	24	16	0	65535	1	-
Cluster_InterfaceVersion	36	4	0	15	1	always "1"

Table 30: Cluster_0_Status - message content (0x600) [ARS 408-21](#)

Signal	Start	Len	Min	Max	Res	Unit
Cluster_NofClustersNear	0	8	0	255	1	-
Cluster_NofClustersFar	8	8	0	255	1	-
Cluster_MeasCounter	24	16	0	65535	1	-
Cluster_InterfaceVersion	36	4	0	15	1	always "1"

Table 31: Cluster_0_Status – signal description (0x600).

Start	Signal	Description
0	Cluster_NofClustersNear	Number of detected clusters in the near range scan (see below)
8	Cluster_NofClustersFar	Number of detected clusters in the far range scan (Near range and far range together max. 250 clusters; in case of activation of the quality mode app. 110 – 120 clusters before the CAN-Bus gets an overrun, which must be avoided) ARS 404-21: this value is not supported
24	Cluster_MeasCounter	Measurement cycle counter (counting up since startup of sensor and restarting at 0 when > 65535)
36	Cluster_InterfaceVersion	Cluster list CAN interface Version-No. . It is always “1” till any Cluster Identifier will be changed in any coming SW-update.

9.2. Cluster general information (0x701)

This message contains the position and velocity of the clusters and is sent repeatedly for all the detected clusters (first near scan, then far scan). Each of the two cluster lists is range sorted. If there are more than 250 clusters, only the first 250 clusters are sent.

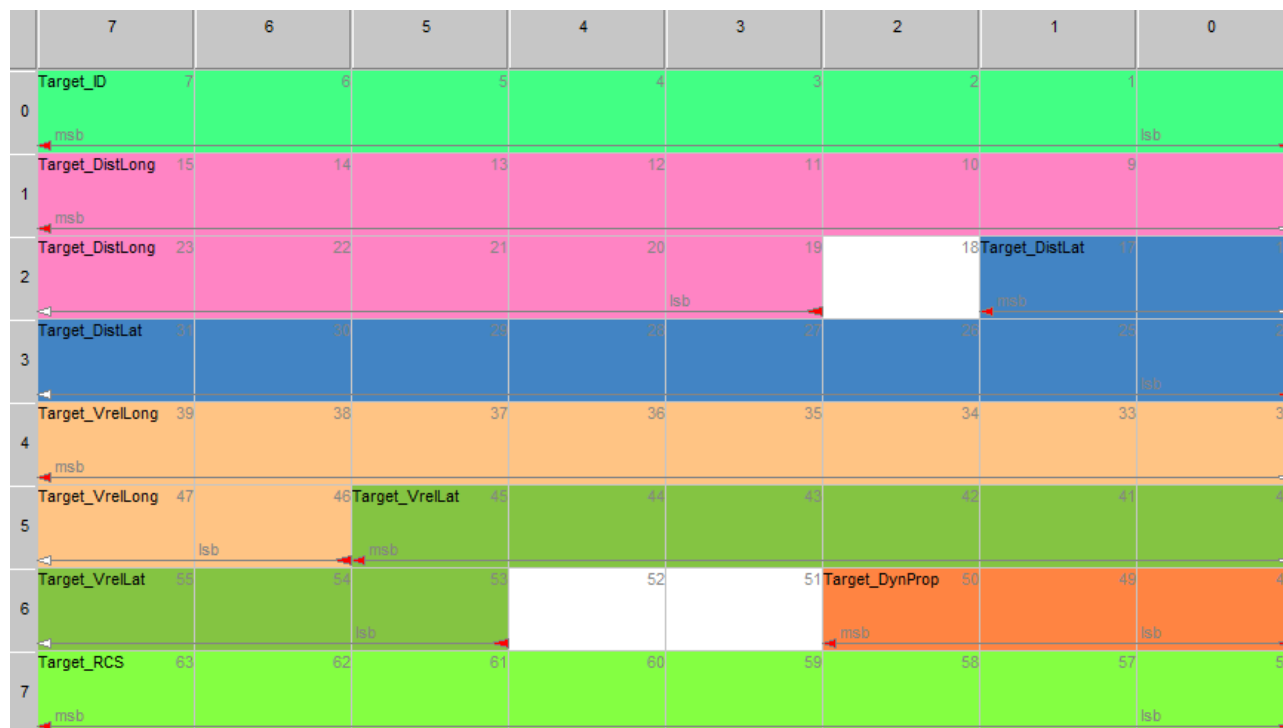


Figure 26: Cluster_1_General - message layout (0x701)

Table 32: Cluster_1_General - message content (0x701)

Signal	Start	Len	Offset	Min	Max	Res	Unit
Cluster_ID	0	8		0	255	1	
Cluster_DistLong	19	13	-500	-500	+1138.2	0.2	m
Cluster_DistLat	24	10	-102.3	-102.3	+102.3	0.2	m
Cluster_VrelLong	46	10	-128.00	-128.00	127.75	0.25	m/s
Cluster_DynProp	48	3		0	7	1	0x0: moving 0x1: stationary 0x2: oncoming 0x3: stationary candidate 0x4: unknown 0x5: crossing stationary 0x6: crossing moving 0x7: stopped

Cluster_VrelLat	53	9	-64.00	-64.00	63.75	0.25	m/s always "0"
Cluster_RCS	56	8	-64.0	-64.0	63.5	0.5	dBm ²

Table 33: Cluster_1_General - signal description (0x701)

Start	Signal	Description
0	Cluster_ID	Cluster number
19	Cluster_DistLong	Longitudinal (x) coordinate
24	Cluster_DistLat	Lateral (y) coordinate
46	Cluster_VrelLong	Relative velocity in longitudinal direction (x)
53	Cluster_VrelLat	Relative velocity in lateral direction (y) This value is permanently set to "Zero"
48	Cluster_DynProp	Dynamic property of cluster to indicate if it is moving or not
56	Cluster_RCS	Radar cross section

9.3. Cluster quality information (0x702)

This message contains the quality information of the clusters and is only sent if it was activated in signal **RadarCfg_SendQuality** (0x200). It is sent repeatedly for all clusters in the same way as message **Cluster_1_General** (0x701). If this message is activated the maximum number of clusters has to be limited to 125 either by filtering or environment.

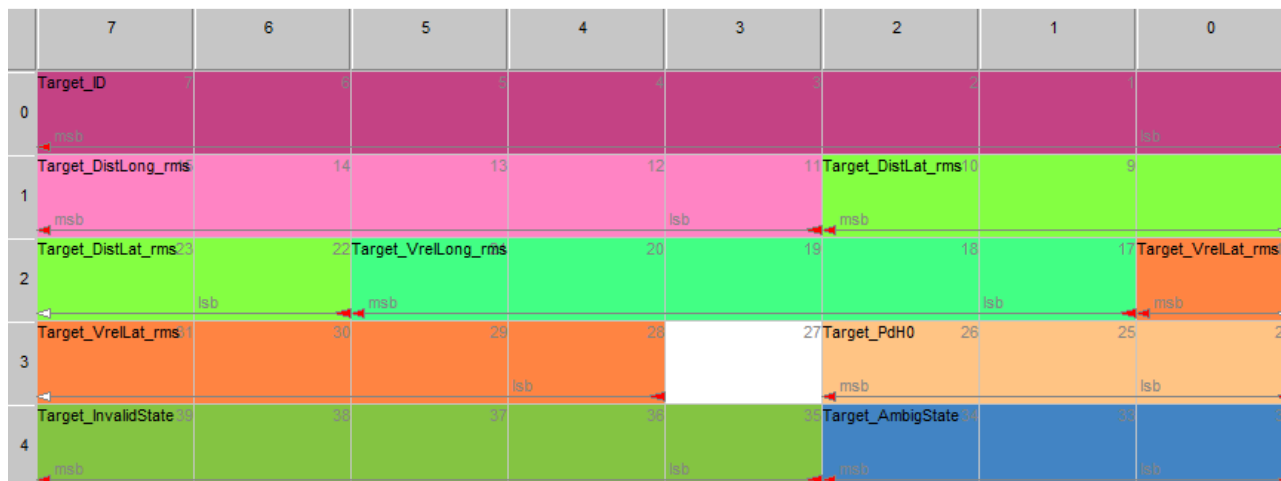


Figure 27: Cluster_2_Quality - message layout (0x702)

Table 34: Cluster_2_Quality - message content (0x702)

Signal	Start	Len	Min	Max	Res	Unit
Cluster_ID	0	8	0	255	1	-
Cluster_DistLong_rms	11	5	0	31	1	see Table 36
Cluster_VrellLong_rms	17	5	0	31	1	see Table 36
Cluster_DistLat_rms	22	5	0	31	1	see Table 36
Cluster_Pdh0	24	3	0	7	1	0x0: invalid 0x1: <25% 0x2: <50% 0x3: <75% 0x4: <90% 0x5: <99% 0x6: <99.9% 0x7: <=100%
Cluster_VrellLat_rms	28	5	0	31	1	see Table 36

Cluster_AmbigState	32	3	0	7	1	0x0: invalid (cannot be used) 0x1: ambiguous (cluster is not clear as there is fuzziness (darkness, ambiguity)). (Should not be used and not recommended anyway, because it could be there twice or even more times) 0x2: staggered ramp (the cluster is not clear somehow as the fuzziness (ambiguity) has been resolved partially). (Should not be used and not recommended anyway) 0x3: unambiguous (everything is clear, the fuzziness(ambiguity) has been resolved – it can be used and recommended for usage) 0x4: stationary candidates (everything is clear and probably there is something stationary (fixed) as the fuzziness(ambiguity) has been resolved – it can be used and recommended for usage)
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Cluster_InvalidState	35	5	0	31	1	0x00: Valid 0x01: Invalid due to low RCS 0x02: Invalid due to near-field artefact 0x03: Invalid far range Cluster because not confirmed in near range 0x04: Valid Cluster with low RCS 0x05: reserved 0x06: Invalid Cluster due to high mirror probability 0x07: Invalid because outside sensor field of view 0x08: Valid Cluster with azimuth correction due to elevation 0x09: Valid Cluster with high child probability 0x0A: Valid Cluster with high probability of being a 50 deg artefact 0x0B: Valid Cluster but no local maximum 0x0C: Valid Cluster with high artefact probability 0x0D: reserved 0x0E: Invalid Cluster because it is a harmonics 0x0F: Valid Cluster above 95 m in near range 0x10: Valid Cluster with high multi-target probability 0x11: Valid Cluster with suspicious angle
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Table 35: Cluster_2_Quality - signal description (0x702)

Start	Signal	Description
0	Cluster_ID	Cluster number
11	Cluster_DistLong_rms	Standard deviation of longitudinal distance
22	Cluster_DistLat_rms	Standard deviation of lateral distance
17	Cluster_VrelLong_rms	Standard deviation of longitudinal relative velocity
28	Cluster_VrelLat_rms	Standard deviation of lateral relative velocity

24	Cluster_Pdh0	False alarm probability of Cluster (i.e. probability for being an artefact caused by multipath or similar)
32	Cluster_AmbigState	State of Doppler (radial velocity) ambiguity solution (see the seventh row in table 34 for more details)
35	Cluster_InvalidState	State of Cluster validity state (see Table 34 for more details)

Table 36: Signal value table for Cluster_DistLong_rms, Cluster_DistLat_rms, Cluster_VrelLong_rms and Cluster_VrelLat_rms (0x702).

Parameter	Value for signals Cluster_DistLong_rms, Cluster_DistLat_rms [m] Cluster_VrelLong_rms, Cluster_VrelLat_rms [m/s]
0x0	<0.005
0x1	<0.006
0x2	<0.008
0x3	<0.011
0x4	<0.014
0x5	<0.018
0x6	<0.023
0x7	<0.029
0x8	<0.038
0x9	<0.049
0xA	<0.063
0xB	<0.081
0xC	<0.105
0xD	<0.135
0xE	<0.174
0xF	<0.224
0x10	<0.288
0x11	<0.371
0x12	<0.478
0x13	<0.616
0x14	<0.794
0x15	<1.023
0x16	<1.317
0x17	<1.697
0x18	<2.187
0x19	<2.817

0x1A	<3.630
0x1B	<4.676
0x1C	<6.025
0x1D	<7.762
0x1E	<10.000
0x1F	invalid